Tutorial 10: PCA and Factor Analysis

ENVX2001 – Applied Statistical Methods

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Semester 1

PCA and Factor Analysis

This tutorial uses football-player performance data from Wilson et al. (2017), *Skill not athleticism predicts individual variation in match performance of soccer players*. The study measured semi-professional football players from the University of Queensland Football Club and asked whether morphology, balance, athleticism and motor skill predicted performance in soccer-tennis and 11-a-side matches.

In the paper, the authors measured:

- **Balance**: average time before losing balance while standing on high-density foam with eyes closed.
- **Athletic performance**: 1500 m speed, wall-squat endurance, jump distance, 40 m sprint speed and agility-course speed.
- **Motor skill**: dribbling speed, juggling ability, volley accuracy, passing accuracy and heading accuracy.

The key biological question is whether complex match performance is better explained by general athletic capacity or football-specific motor skill.

We are going to use principal components analysis and factor analysis to interpret the multivariate structure of these player performance traits.

Setup

Load these packages to make the plots look better:

CODE

```
library(EFAtools)
library(ggplot2)
library(ggfortify)
```

Load the data

First open your dataset. The CSV file should be saved in the same folder as this `.qmd` file.

CODE

```
players <- read.csv("data/football_player_performance.csv", header = TRUE)
```

Check the variable names:

CODE

```
names(players)
```

OUTPUT

```
 [1] "player"  "balance" "X1500m"  "squat"   "jump"    "sprint"  "agility"
 [8] "drib"    "jugg"    "voll"    "pass"    "head"
```

Because R changes column names that begin with a number, `1500m` may appear as `X1500m`.

CODE

```
head(players)
```

OUTPUT

```
  player balance   X1500m squat jump   sprint  agility     drib     jugg
1      1   6.485 4.297994   361 2.67 7.272727 3.601286 2.545455 14.80000
```

```
2      2  25.000 4.602194   100 2.55 7.214200 3.530339 2.554162 26.50000
3      3   3.920 3.916449    87 2.51 6.430868 3.412034 2.471042 10.11111
4      4  20.000 3.605769    85 2.36 6.700168 3.227666 2.175814 14.50000
5      5  46.700 4.065041   141 2.52 6.920415 3.446154 2.471042  8.40000
6      6  60.000 3.778338   172 2.53 6.666667 3.530339 2.621416 21.15385
      voll     pass     head
1 11.76471 20.40816 15.15152
2 11.00000 31.40816 14.49275
3 11.76471 17.85714 14.49275
4 12.80855 18.87122 13.70566
5 10.73989 21.81725 16.55008
6 11.62791 24.39024 15.38462
```

CODE

```r
str(players)
```

OUTPUT

```
'data.frame':	24 obs. of  12 variables:
 $ player : int  1 2 3 4 5 6 7 8 9 10 ...
 $ balance: num  6.49 25 3.92 20 46.7 ...
 $ X1500m : num  4.3 4.6 3.92 3.61 4.07 ...
 $ squat  : int  361 100 87 85 141 172 80 216 81 189 ...
 $ jump   : num  2.67 2.55 2.51 2.36 2.52 2.53 2.19 2.38 2.42 2.72 ...
 $ sprint : num  7.27 7.21 6.43 6.7 6.92 ...
 $ agility: num  3.6 3.53 3.41 3.23 3.45 ...
 $ drib   : num  2.55 2.55 2.47 2.18 2.47 ...
 $ jugg   : num  14.8 26.5 10.1 14.5 8.4 ...
 $ voll   : num  11.8 11 11.8 12.8 10.7 ...
 $ pass   : num  20.4 31.4 17.9 18.9 21.8 ...
 $ head   : num  15.2 14.5 14.5 13.7 16.6 ...
```

Description of the variables

CODE

```r
summary(players)
```

OUTPUT

```
     player         balance          X1500m          squat       
 Min.   : 1.00   Min.   : 3.92   Min.   :3.606   Min.   : 79.00  
 1st Qu.: 6.75   1st Qu.:12.00   1st Qu.:3.940   1st Qu.: 84.75  
 Median :12.50   Median :22.92   Median :4.364   Median :103.00  
 Mean   :12.50   Mean   :25.48   Mean   :4.283   Mean   :131.50  
 3rd Qu.:18.25   3rd Qu.:34.99   3rd Qu.:4.549   3rd Qu.:166.00  
 Max.   :24.00   Max.   :60.00   Max.   :4.854   Max.   :361.00  
      jump           sprint         agility           drib      
 Min.   :2.050   Min.   :6.231   Min.   :3.228   Min.   :2.176  
 1st Qu.:2.360   1st Qu.:6.650   1st Qu.:3.379   1st Qu.:2.504  
 Median :2.465   Median :6.932   Median :3.497   Median :2.548  
 Mean   :2.439   Mean   :6.889   Mean   :3.466   Mean   :2.553  
 3rd Qu.:2.535   3rd Qu.:7.175   3rd Qu.:3.545   3rd Qu.:2.632  
 Max.   :2.720   Max.   :7.491   Max.   :3.666   Max.   :2.784  
      jugg             voll             pass            head       
 Min.   : 7.842   Min.   : 7.804   Min.   :10.81   Min.   : 4.182  
 1st Qu.:10.659   1st Qu.: 9.501   1st Qu.:20.41   1st Qu.:14.296  
 Median :14.082   Median :11.250   Median :28.67   Median :15.268  
 Mean   :15.900   Mean   :11.017   Mean   :29.77   Mean   :14.973  
```

```
 3rd Qu.:18.990   3rd Qu.:12.544   3rd Qu.:38.00   3rd Qu.:17.712  
 Max.   :39.857   Max.   :13.889   Max.   :50.00   Max.   :19.231  
```

The variables are:

- `balance`: static balance score
- `X1500m`: speed over 1500 m
- `squat`: wall-squat endurance time
- `jump`: standing jump distance
- `sprint`: 40 m sprint speed
- `agility`: speed through an agility course
- `drib`: dribbling speed through an agility course
- `jugg`: juggling or keep-up ability
- `voll`: volley-kick accuracy
- `pass`: passing accuracy
- `head`: heading accuracy

Higher values generally indicate better performance on that trait.

Correlation matrix

Let's first do a correlation matrix. We exclude the player ID column and analyse the performance traits only.

CODE

```r
cor(players[, 2:12])
```

OUTPUT

```
             balance       X1500m        squat         jump      sprint
balance  1.000000000 -0.120666590  0.056071952  0.005165273  0.14975877
X1500m  -0.120666590  1.000000000  0.159548821  0.096566292  0.34226606
squat    0.056071952  0.159548821  1.000000000  0.432157704  0.30883252
jump     0.005165273  0.096566292  0.432157704  1.000000000  0.33871992
sprint   0.149758766  0.342266059  0.308832520  0.338719924  1.00000000
agility -0.229848417  0.168169479  0.445175401  0.571266808  0.16286933
drib     0.470845996  0.356356256  0.159653733  0.154769848  0.29294354
jugg     0.187921370  0.155839614  0.004928929  0.280997027 -0.07650198
voll     0.397624644  0.006100075  0.153281966  0.025642532  0.32241412
pass     0.416714230  0.155833542 -0.042274047 -0.006327859  0.25026128
head    -0.006255931  0.032500939  0.122024732  0.081488984 -0.10767999
            agility       drib         jugg         voll         pass
balance -0.22984842 0.47084600  0.187921370  0.397624644  0.416714230
X1500m   0.16816948 0.35635626  0.155839614  0.006100075  0.155833542
squat    0.44517540 0.15965373  0.004928929  0.153281966 -0.042274047
jump     0.57126681 0.15476985  0.280997027  0.025642532 -0.006327859
sprint   0.16286933 0.29294354 -0.076501981  0.322414125  0.250261285
agility  1.00000000 0.02135601  0.185489117 -0.303537885 -0.015220367
drib     0.02135601 1.00000000  0.346830908  0.313925571  0.507806703
jugg     0.18548912 0.34683091  1.000000000  0.107128520  0.112221815
voll    -0.30353789 0.31392557  0.107128520  1.000000000  0.530011484
pass    -0.01522037 0.50780670  0.112221815  0.530011484  1.000000000
```

```
head     0.36781797 0.07426933  0.235577420  0.394861742  0.341212829
                head
balance -0.006255931
X1500m   0.032500939
squat    0.122024732
jump     0.081488984
sprint  -0.107679989
agility  0.367817975
drib     0.074269329
jugg     0.235577420
voll     0.394861742
pass     0.341212829
head     1.000000000
```

CODE

```
Corrmatrix <- cor(players[, 2:12])
```

A correlation matrix lets us see which traits tend to vary together. For example, if several skill traits are positively correlated, this suggests that some players have generally high technical ability across multiple football-specific tasks.

Bartlett's Test of Sphericity

Now we can do Bartlett's Test of Sphericity. This test compares the correlation matrix to an identity matrix. If it is significant, it is worth doing a PCA.

CODE

```
BARTLETT(Corrmatrix, N = nrow(players), cor_method = c("pearson"))
```

OUTPUT

```
✓ The Bartlett's test of sphericity was significant at an alpha level of .05.
  These data are probably suitable for factor analysis.

  χ²(55) = 81.19, p = 0.012
```

A significant result means the variables are sufficiently correlated for a dimension-reduction method such as PCA or factor analysis.

Principal Components Analysis

Note: to make this a PCA based on a correlation matrix, we have to scale the variables, hence `scale = TRUE`. There are two main principal components functions, but they are very similar. Note `prcomp` calls the loadings "rotations", not to be confused with rotations below.

CODE

```
pca1 <- prcomp(players[, 2:12], scale = TRUE)
pca2 <- princomp(players[, 2:12], cor = TRUE)
```

CODE

```
summary(pca1)
```

OUTPUT

```
Importance of components:
                          PC1    PC2    PC3    PC4     PC5     PC6     PC7
Standard deviation     1.6999 1.4767 1.1795 1.0726 1.02453 0.80985 0.79892
Proportion of Variance 0.2627 0.1982 0.1265 0.1046 0.09542 0.05962 0.05803
Cumulative Proportion  0.2627 0.4609 0.5874 0.6920 0.78742 0.84704 0.90507
                           PC8     PC9    PC10    PC11
Standard deviation     0.60575 0.55566 0.52948 0.29702
Proportion of Variance 0.03336 0.02807 0.02549 0.00802
Cumulative Proportion  0.93842 0.96649 0.99198 1.00000
```

CODE

```
summary(pca2)
```

OUTPUT

```
Importance of components:
                          Comp.1    Comp.2    Comp.3    Comp.4     Comp.5
Standard deviation     1.6999027 1.4766764 1.1794865 1.0726210 1.02452713
Proportion of Variance 0.2626972 0.1982339 0.1264717 0.1045924 0.09542326
Cumulative Proportion  0.2626972 0.4609311 0.5874028 0.6919951 0.78741840
                           Comp.6    Comp.7    Comp.8     Comp.9    Comp.10
Standard deviation     0.80984729 0.7989246 0.6057459 0.55566144 0.52948396
Proportion of Variance 0.05962297 0.0580255 0.0333571 0.02806906 0.02548666
Cumulative Proportion  0.84704137 0.9050669 0.9384240 0.96649302 0.99197968
                           Comp.11
Standard deviation     0.297024364
Proportion of Variance 0.008020316
Cumulative Proportion  1.000000000
```

The first principal component describes the major axis of variation among players. In the Wilson et al. paper, separate PCAs were used to summarise athletic performance and motor skill. Here, we are doing one combined PCA across balance, athletic performance and motor skill traits, so the first few components may separate general performance, athleticism and football-specific skill.

Loadings

Let's look at the loadings. Called "rotations" in `prcomp` and "loadings" in `princomp`. They are the Pearson's correlation between that variable and that Principal Component.

CODE

```
pca1
```

OUTPUT

```
Standard deviations (1, .., p=11):
 [1] 1.6999027 1.4766764 1.1794865 1.0726210 1.0245271 0.8098473 0.7989246
 [8] 0.6057459 0.5556614 0.5294840 0.2970244

Rotation (n x k) = (11 x 11):
              PC1         PC2         PC3         PC4         PC5         PC6
```

```
balance 0.2936386 -0.35087385 -0.05458821 -0.01236935 -0.50256454  0.22417684
X1500m  0.2299058  0.15552197 -0.28197657 -0.42123014  0.58080664 -0.18787938
squat   0.2547779  0.35550940 -0.14404714  0.34331707 -0.16954618 -0.19781100
jump    0.2656109  0.43501720 -0.01783357  0.04579993 -0.35433245 -0.03857511
sprint  0.3261362  0.10208432 -0.52353137  0.20056752  0.10706161 -0.05512777
agility 0.1673372  0.55847807  0.21211288  0.02473830  0.03452894  0.42295776
drib    0.4327319 -0.14166176 -0.12392103 -0.36532953 -0.05741543  0.19918699
jugg    0.2447494  0.06217972  0.37349976 -0.55564980 -0.25869776 -0.42767336
voll    0.3619836 -0.32312886  0.07293446  0.39291316  0.07224943 -0.49119815
pass    0.3945243 -0.28979483  0.10341401  0.06052617  0.23371625  0.47674183
head    0.2392728  0.04926387  0.63756754  0.24769351  0.33626830 -0.02753477
                PC7         PC8          PC9        PC10        PC11
balance  0.20538168 -0.54118171  0.365326098 -0.12428219 -0.02713210
X1500m   0.18718969 -0.15409295  0.479562718 -0.01307712 -0.06663040
squat    0.65671346  0.09650932 -0.081047782  0.33932493  0.19978295
jump    -0.42236751  0.37267671  0.478977025 -0.18468479  0.17304917
sprint  -0.42420682 -0.39669448 -0.404284981 -0.05135179  0.22582954
agility -0.01039040 -0.22522259 -0.142378623  0.02285589 -0.60231545
drib     0.25091726  0.42491622 -0.384734195 -0.45697622 -0.02280376
jugg    -0.15523189 -0.17263458 -0.237654632  0.35303879  0.05601670
voll    -0.10513326  0.15078192  0.048968103 -0.04463885 -0.56584548
pass    -0.17259657  0.22907881  0.099702647  0.59253599  0.14422004
head     0.07178534 -0.20375286 -0.003474513 -0.38133775  0.40809767
```

CODE

```
loadings(pca2)
```

OUTPUT

```

Loadings:
        Comp.1 Comp.2 Comp.3 Comp.4 Comp.5 Comp.6 Comp.7 Comp.8 Comp.9 Comp.10
balance  0.294  0.351                0.503  0.224  0.205  0.541  0.365  0.124
X1500m   0.230 -0.156  0.282  0.421 -0.581 -0.188  0.187  0.154  0.480
squat    0.255 -0.356  0.144 -0.343  0.170 -0.198  0.657               -0.339
jump     0.266 -0.435                0.354        -0.422 -0.373  0.479  0.185
sprint   0.326 -0.102  0.524 -0.201 -0.107        -0.424  0.397 -0.404
agility  0.167 -0.558 -0.212                0.423         0.225 -0.142
drib     0.433  0.142  0.124  0.365         0.199  0.251 -0.425 -0.385  0.457
jugg     0.245        -0.373  0.556  0.259 -0.428 -0.155  0.173 -0.238 -0.353
voll     0.362  0.323        -0.393        -0.491 -0.105 -0.151
pass     0.395  0.290 -0.103        -0.234  0.477 -0.173 -0.229        -0.593
head     0.239        -0.638 -0.248 -0.336                0.204         0.381
        Comp.11
balance
X1500m
squat   -0.200
jump    -0.173
sprint  -0.226
agility  0.602
drib
jugg
voll     0.566
pass    -0.144
head    -0.408

               Comp.1 Comp.2 Comp.3 Comp.4 Comp.5 Comp.6 Comp.7 Comp.8 Comp.9
SS loadings     1.000  1.000  1.000  1.000  1.000  1.000  1.000  1.000  1.000
Proportion Var  0.091  0.091  0.091  0.091  0.091  0.091  0.091  0.091  0.091
Cumulative Var  0.091  0.182  0.273  0.364  0.455  0.545  0.636  0.727  0.818
               Comp.10 Comp.11
SS loadings      1.000   1.000
```

```
Proportion Var    0.091    0.091
Cumulative Var    0.909    1.000
```

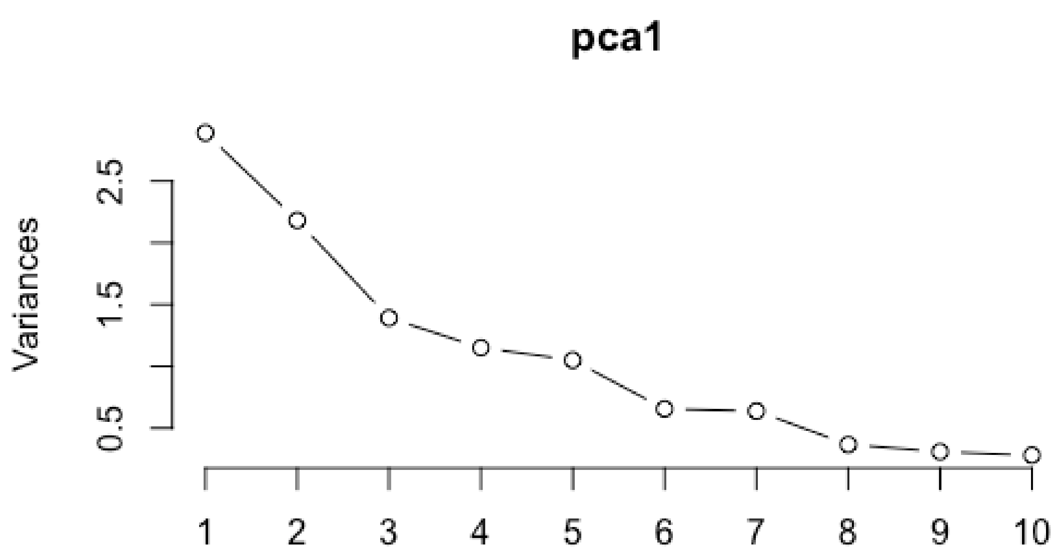
Interpretation:

- Variables with large positive or negative loadings contribute strongly to that component.
- If all traits load in the same direction on PC1, PC1 can be interpreted as general player performance.
- If athletic traits load in one direction and skill traits load in another, the component may represent a contrast between athletic capacity and football-specific skill.

Screeplot

To do a screeplot, follow the commands below. Note this is the standard deviations, which are just square root of the variances or eigenvalues.

```
CODE
screeplot(pca1, type = "lines")
```



The screeplot helps decide how many components are worth interpreting. A common rule of thumb is to keep components before the curve flattens out.

Principal Component Scores

To get your principal components scores for plotting and analysis, do the following:

CODE

```
pca1$x
```

OUTPUT

```
             PC1         PC2         PC3          PC4         PC5         PC6
 [1,]  1.22265294  3.21272139 -0.84513278  1.729875647 -0.41800665 -1.16536979
 [2,]  1.01549076  0.72909627 -0.07487739 -1.126299782  0.08030236 -0.44330634
 [3,] -1.95431230  0.13616424  0.62019420  0.683133943 -0.08692596 -0.65692178
 [4,] -2.74171773 -1.49202883  0.50124852  1.735454370 -0.78052160 -1.85418850
 [5,] -0.29218316 -0.08747326 -0.11988125  1.108831773 -0.99391905  0.32842292
 [6,]  0.91997987 -0.28734416  0.84641464  0.232906297 -2.57193000  0.29232622
 [7,] -0.95858539 -2.60131264  1.43635426  0.990763299  0.22009132  0.24168604
 [8,]  0.04244428  0.86220012 -1.13368412  0.678632536 -0.23885615  0.30250751
 [9,] -1.12244287 -0.06955280  1.12603291  0.288229193 -0.12664793  1.91010162
[10,]  3.39686967  0.83136647 -0.43475555  1.121845842  0.67921147  0.65544911
[11,]  1.10678403 -2.25184331 -2.80892930 -1.372109965 -1.76088086  0.26168616
[12,]  1.62463419 -0.31440598 -0.62813600 -0.915492585 -0.10688046 -1.21576231
[13,] -0.57673686  1.50371749  1.09522239  0.357669243 -0.40536591  1.07877742
[14,] -2.45940301  1.55096778 -1.73104596 -0.815402453  0.44465656 -0.23066431
[15,] -1.83385997 -1.38084282 -0.43730279 -0.678128599  0.95811502  0.19042700
[16,]  0.42353253 -0.66991754  0.33476898  1.159555934  1.82618886  0.02417214
[17,]  2.33632926 -2.16227916 -0.18130741 -0.003407567  0.72452040  0.03044801
[18,] -1.18106445  1.01946486 -1.42004479  0.469515124 -0.72656471  0.99789886
[19,] -0.28781540  1.77196130  0.32939063 -1.408868336  0.66689434 -0.33260253
[20,]  0.77348049  0.82720824  3.03423799 -2.239773715 -1.30000966 -0.51164628
[21,]  2.10946365 -1.29318452  0.20187351  0.034487363  0.57647043  0.44954616
[22,] -3.17285972 -0.60941350 -0.65733911 -1.356394009  0.79164937  0.24990229
[23,]  0.16649597  1.91778410  0.60759087 -0.489210312  1.41139229  0.25681745
[24,]  1.44282322 -1.14305374  0.33910755 -0.185813240  1.13701651 -0.85970707
             PC7         PC8          PC9        PC10        PC11
 [1,]  1.06626147  0.74992918 -0.609228418  0.46483136  0.17305201
 [2,] -1.09718405 -0.55422064  0.020706681  0.30612703 -0.11318983
 [3,] -0.48184786  1.35102821  0.176215524 -0.67519862 -0.43378351
 [4,] -1.06403599 -0.28180079  0.378695770  0.62881031  0.21732729
 [5,]  0.19619194 -0.83106823  0.854657258 -0.89586615  0.36499683
 [6,]  0.70036296 -0.63152387  0.184658562 -0.46021429 -0.40732572
 [7,]  0.47708118  0.29020781 -0.199234399 -0.10259401  0.08118763
 [8,]  1.15124813 -0.95643173 -0.276070905 -0.34757651  0.10353943
 [9,] -0.72084050  0.16806957  0.038709675  1.07069258  0.30262699
[10,] -0.65278915  0.16655620  0.932371171  0.05236096 -0.10214435
[11,] -0.12849364  0.55873899  0.297675862  0.89339944 -0.12124663
[12,]  0.78971314  0.03142220 -0.228192952  0.24684205  0.39712028
[13,]  0.84348491  0.35975295 -0.505277549  0.58132253 -0.50547940
[14,] -0.33609876 -0.32114157  0.801087448 -0.11434646 -0.54458835
[15,]  0.95117438 -0.59904114 -0.831463657  0.01884134 -0.16940143
[16,]  0.03885635 -1.02979447 -0.006515691  0.66914191 -0.25696656
[17,] -0.25250279 -0.04466779 -0.360784257 -0.27940159  0.17394913
[18,] -1.68675889  0.24054300 -1.225329831 -0.62610765  0.31313624
[19,] -0.98192696 -0.61152963 -0.431957026 -0.35399954  0.06783876
[20,] -0.05633572 -0.03278367 -0.008561725  0.05457287  0.13508982
[21,]  0.02130256  0.74598418  0.172521694 -0.78114525 -0.03614367
[22,]  1.20492210  0.59954232  0.449245111 -0.24242268  0.33615101
[23,]  0.36131351  0.36550291  0.878867693  0.12037610  0.36393275
[24,] -0.34309832  0.26672601 -0.502796039 -0.22844571 -0.33967872
```

So to combine them with the original variables, do this:

CODE

```
playerpcscores <- cbind(players, pca1$x)
```

Each player now has a score on each principal component. These scores tell us where each player sits on the multivariate axes of football performance.

Biplot

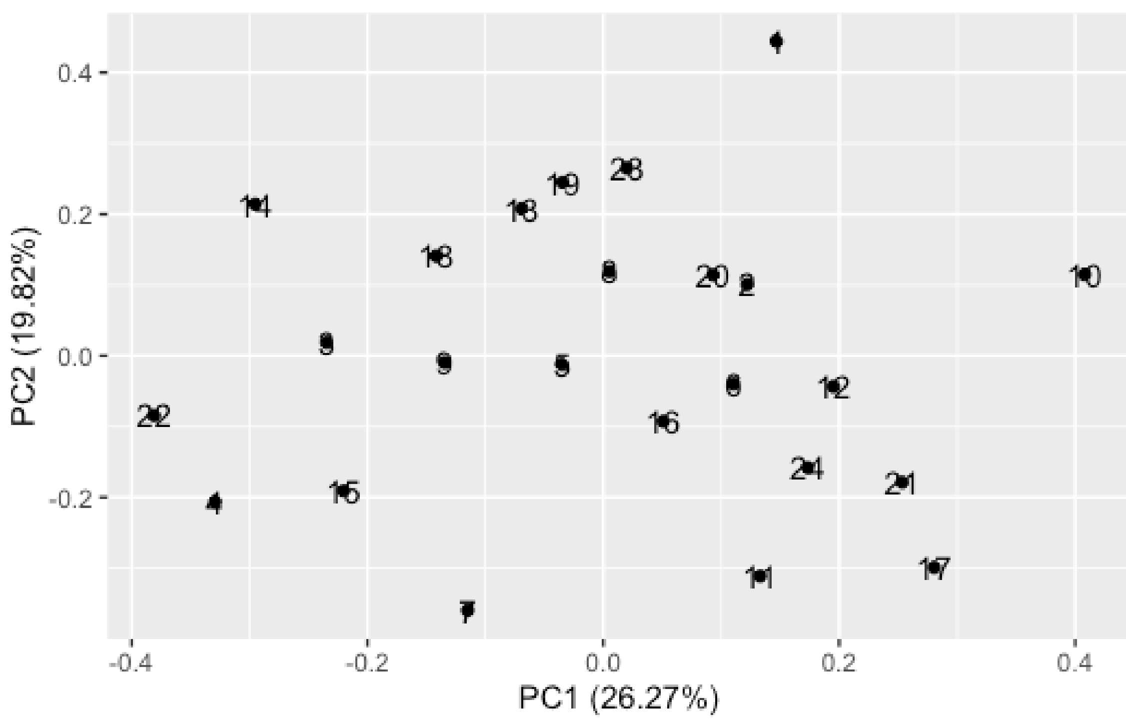
We can produce a biplot in ggplot2. Here the player number is used as a label.

CODE

```
autoplot(pca1, data = playerpcscores, label = TRUE, label.label = "player")
```

OUTPUT

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
ℹ Please use tidy evaluation idioms with `aes()`.
ℹ See also `vignette("ggplot2-in-packages")` for more information.
ℹ The deprecated feature was likely used in the ggfortify package.
  Please report the issue at <https://github.com/sinhrks/ggfortify/issues>.
```



We could add the loadings, but they can be a bit messy. It is often best to look at the loadings or rotations tables directly.

You can make the plot using different symbols, change background etc through reading the ggplot2 documentation.

Regression using PC scores

In the lecturer dataset, we used ANOVA to test whether PC scores differed among groups. This football dataset does not include grouping variables such as team grade or playing position, so here we use a regression example instead.

For example, we can ask whether PC1 is associated with juggling ability:

CODE

```
LM1 <- lm(PC1 ~ jugg, data = playerpcscores)
summary(LM1)
```

OUTPUT

```

Call:
lm(formula = PC1 ~ jugg, data = playerpcscores)

Residuals:
    Min      1Q  Median      3Q     Max
-2.6088 -1.1795  0.1164  0.8217  3.7469

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.51001    0.77408  -1.951   0.0639 .
jugg         0.09497    0.04425   2.146   0.0432 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.581 on 22 degrees of freedom
Multiple R-squared:  0.1731,    Adjusted R-squared:  0.1355
F-statistic: 4.605 on 1 and 22 DF,  p-value: 0.04316
```

This is only an example. In a full analysis, you might use PC scores as predictors of match success, soccer-tennis score or network centrality.

Scatterplot

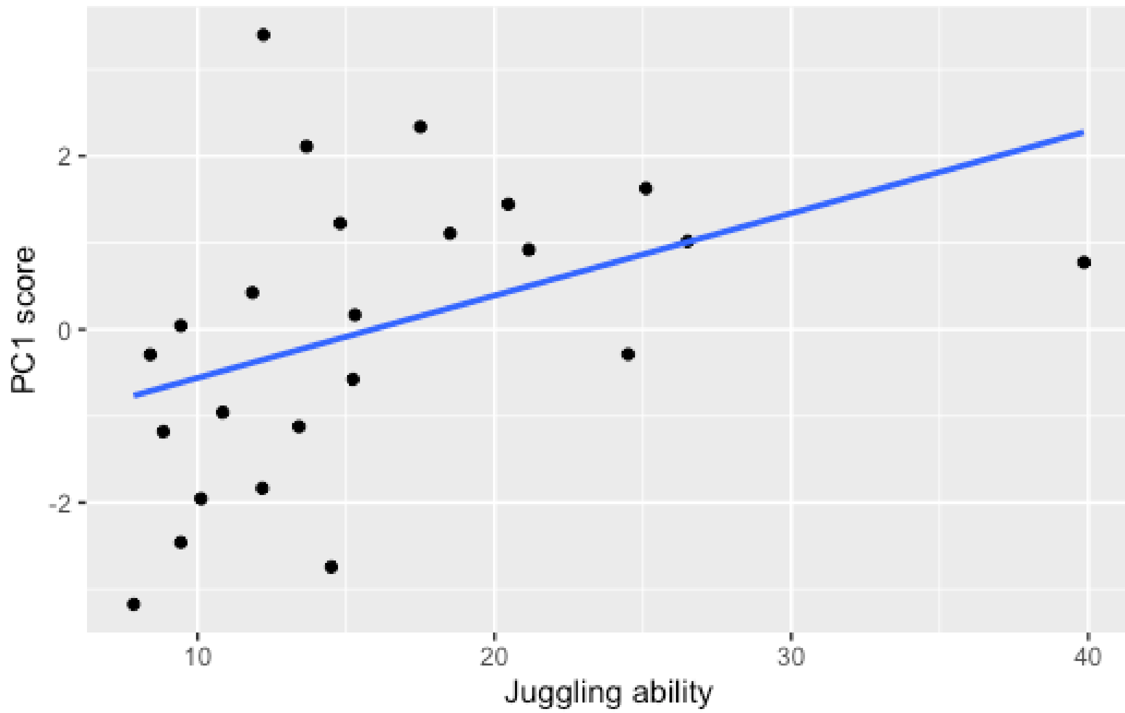
To do a scatterplot of variables use ggplot2:

CODE

```
ggplot(playerpcscores, aes(x = jugg, y = PC1)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x = "Juggling ability", y = "PC1 score")
```

OUTPUT

```
`geom_smooth()` using formula = 'y ~ x'
```



Factor Analysis with Varimax Rotation

To do a Varimax rotation on your principal components, follow these commands. Do a factor analysis with the rotation being Varimax.

Here we start with three factors because the paper describes three broad biological domains: balance, athletic performance and motor skill.

CODE

```
fa1 <- factanal(players[, 2:12], 3, rotation = "varimax")
fa1
```

OUTPUT

```

Call:
factanal(x = players[, 2:12], factors = 3, rotation = "varimax")

Uniquenesses:
balance  X1500m   squat    jump  sprint agility    drib    jugg    voll    pass
  0.722   0.920   0.656   0.524   0.504   0.005   0.706   0.913   0.171   0.607
   head
  0.005

Loadings:
        Factor1 Factor2 Factor3
balance  0.524
X1500m   0.105   0.261
squat    0.112   0.567   0.102
```

```
jump              0.687
sprint   0.441    0.525   -0.158
agility -0.444    0.801    0.395
drib     0.468    0.273
jugg              0.168    0.226
voll     0.849             0.322
pass     0.538    0.133    0.292
head                       0.993

               Factor1 Factor2 Factor3
SS loadings      1.935   1.905   1.427
Proportion Var   0.176   0.173   0.130
Cumulative Var   0.176   0.349   0.479

Test of the hypothesis that 3 factors are sufficient.
The chi square statistic is 19.84 on 25 degrees of freedom.
The p-value is 0.755
```

The rotated factor loadings may help identify clusters of variables. For example:

- one factor may represent athletic performance
- one factor may represent football-specific motor skill
- one factor may represent balance or a narrower technical component

Try changing the number of factors and compare whether the interpretation becomes clearer or less clear:

CODE

```r
fa2 <- factanal(players[, 2:12], 2, rotation = "varimax")
fa2
```

OUTPUT

```

Call:
factanal(x = players[, 2:12], factors = 2, rotation = "varimax")

Uniquenesses:
balance  X1500m   squat    jump  sprint agility    drib    jugg    voll    pass
  0.669   0.932   0.747   0.633   0.805   0.005   0.629   0.894   0.359   0.497
   head
  0.712

Loadings:
        Factor1 Factor2
balance  0.573
X1500m   0.133   0.224
squat            0.497
jump             0.606
sprint   0.336   0.286
agility -0.320   0.945
drib     0.570   0.216
jugg     0.194   0.262
voll     0.799
pass     0.677   0.212
head     0.253   0.473

               Factor1 Factor2
SS loadings       2.09   2.027
Proportion Var    0.19   0.184
Cumulative Var    0.19   0.374
```

```
Test of the hypothesis that 2 factors are sufficient.
The chi square statistic is 33.83 on 34 degrees of freedom.
The p-value is 0.476
```

CODE

```r
fa4 <- factanal(players[, 2:12], 4, rotation = "varimax")
fa4
```

OUTPUT

```

Call:
factanal(x = players[, 2:12], factors = 4, rotation = "varimax")

Uniquenesses:
balance  X1500m   squat    jump  sprint agility    drib    jugg    voll    pass 
  0.677   0.839   0.618   0.511   0.521   0.005   0.005   0.815   0.005   0.555 
   head 
  0.101 

Loadings:
        Factor1 Factor2 Factor3 Factor4
balance  0.423           0.367         
X1500m           0.167   0.364         
squat            0.602                 
jump             0.686   0.117         
sprint   0.338   0.541   0.150  -0.220 
agility -0.485   0.747   0.116   0.434 
drib     0.291           0.950         
jugg                     0.353   0.229 
voll     0.966                   0.235 
pass     0.465           0.394   0.264 
head     0.176                   0.929 

               Factor1 Factor2 Factor3 Factor4
SS loadings      1.804   1.742   1.510   1.293
Proportion Var   0.164   0.158   0.137   0.118
Cumulative Var   0.164   0.322   0.460   0.577

Test of the hypothesis that 4 factors are sufficient.
The chi square statistic is 9.89 on 17 degrees of freedom.
The p-value is 0.908
```

Final interpretation

Wilson et al. found that motor skill was a stronger predictor of soccer-tennis and 11-a-side match performance than general athleticism. In this tutorial, PCA and factor analysis are used to explore the same biological idea from the trait data: whether football players vary mostly along a general performance axis, or whether athletic and technical skill traits form separate multivariate dimensions.